

Deep Learning Project

Sentiment Analysis

Group #2

Aitua Omigie

Danylo Gula

Esther Mbugua

Leshawn Phillips

Safwan Hossain

Tatsuya Nakano

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Background & Problem Statement

Builds a sentiment analysis model that accurately classifies financial news sentences into three sentiment categories. The model should provide actionable insights for investors and financial institutions by identifying market sentiment trends.

Our Approach

1. Understand Business Objective
2. Analyze the dataset
3. Build the model
4. Design deep architecture
5. Handle imbalances
6. Tune and evaluate



The Dataset



**10,687 Financial News Sentences
labeled with sentiment. The
distribution of sentiment labels is
as follows:**

- Neutral: 6,008 (56.2%)
- Positive: 3,215 (30.1%)
- Negative: 1,464 (13.7%)

Challenges

Class Imbalance

- Favours the "neutral" class negatively affecting the model's performance
- Class weighting is used in our loss function to reduce this impact

Contextual Nuance

- Financial language contains a lot of nuance and required context therefore we will use more complex preprocessor and attention layer

Data Preprocessing

- **Stopword Removal:** Removes common words that do not contribute to sentiment
- **De-Noising:** Removing noisy characters while keeping certain symbols (\$,%).
- **Filtering:** Find tickers (AAPL, GOOG, NVDA), de-noise, keep them.
- **Tokenization:** Split sentences into individual words.
- **Lemmatization:** Reduce words to their base form, skip idioms.

The Model

Bidirectional LSTM with an attention mechanism to capture contextual information and focus on relevant words.

The Architecture:

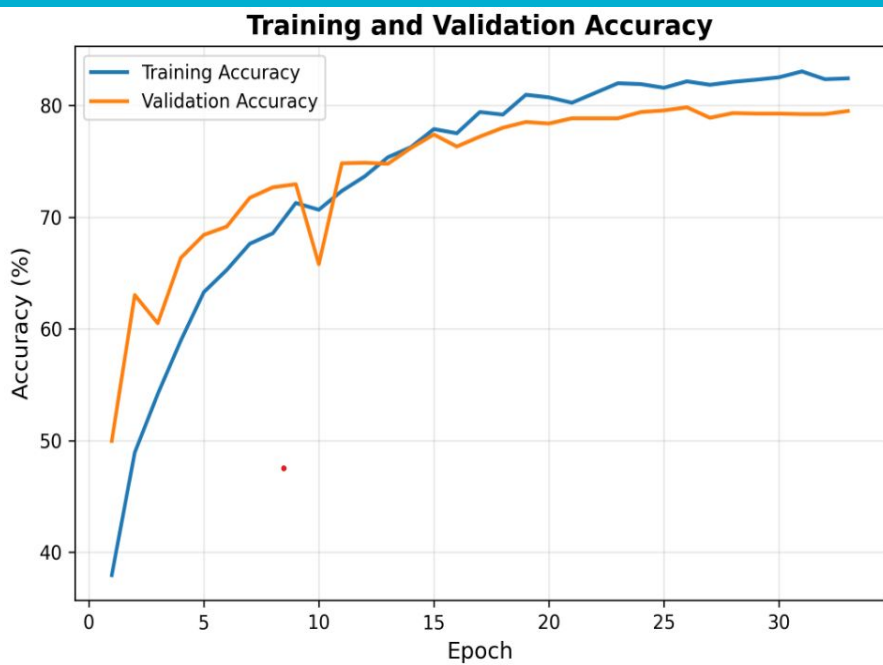
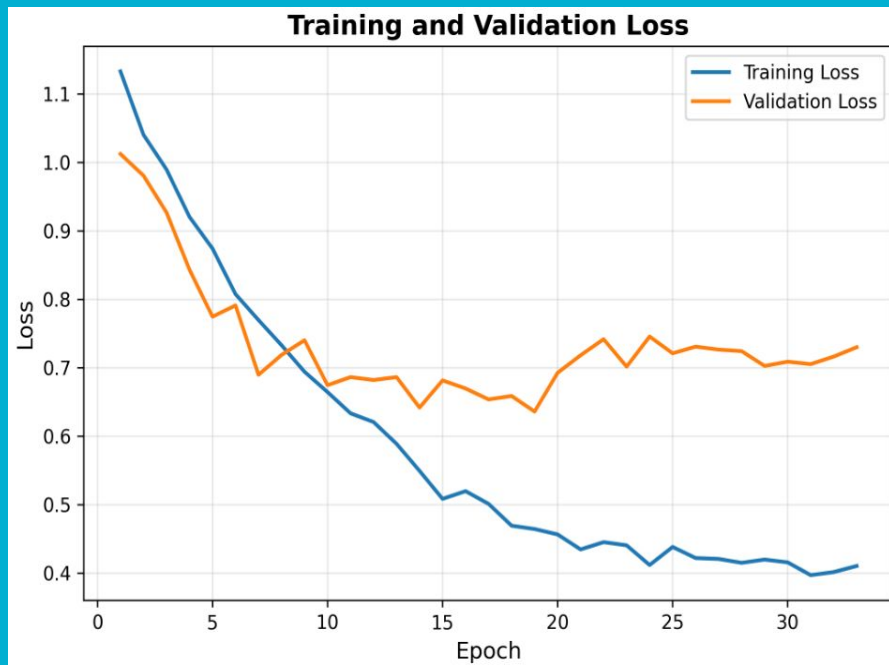
- **Embedding Layer:** Converts words into dense vectors.
 - **LSTM Layer:** Captures sequential dependencies in the text.
 - **Attention Layer:** Focuses on important words in the sentence.
 - **Regularization & Normalization**
 - **Classifier Layer:** Maps combined LSTM and Attention features to sentiments.
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Hyperparameters

- Embed Dim: 192
- Hidden Dim: 128
- Number of Layers: 3
- Dropout Rate: 0.5
- Learning Rate: 0.001
- Batch Size: 64

```
# hyperparameter grid
param_grid = {
    'embedding_dim': [128, 192, 256], # usually favours 192
    'hidden_dim': [64, 128, 192], # 128 almost always chosen
    'n_layers': [3], # no more or it tunes for hours
    'dropout': [0.3, 0.5], # usually favours 0.5
    'lr': [0.001], # 0.001 almost always chosen
    'batch_size': [64] # 64 almost always chosen
}
```

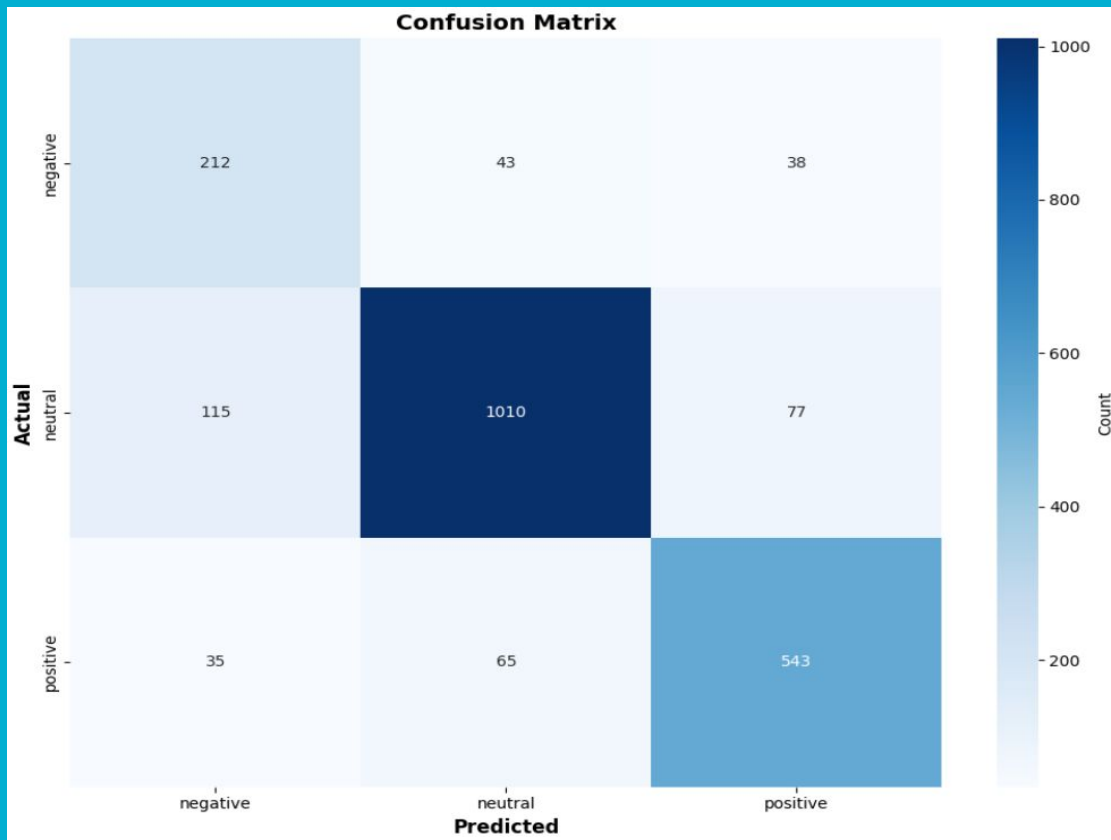
Training & Validation



Confusion Matrix

Predicted vs Actual:

- 82.55% accuracy
- 17.45% error rate



Confusion Matrix

Class Weights:

- Positive: 1.1080
- Neutral: 0.5929
- Negative: 2.4343



Recall:

- Positive: 0.85
- Neutral: 0.81
- Negative: 0.82

Precision:

- Positive: 0.82
- Neutral: 0.92
- Negative: 0.58

Conclusion

Results:

We developed a well-performing LSTM to classify the sentiments of Financial News Headlines to predict whether a headline represents negative, positive, or neutral news sentiment.

Next Steps:

- Explore transformer models like BERT for deeper context
- Apply data augmentation for negative class
- Experiment with focal loss for imbalance

Conclusion

Possible Applications:

- Input other ML Models
 - News Headline Aggregators
- Stock Predictor
 - Gauge market sentiments to inform buy/sell decisions
- Customer Satisfaction Tools
- Mental Health Tools
 - Detect tone shifts in patient journals or session transcripts



Demo + Try Model Out for Yourself!

finance-sentiment.streamlit.app/

